Technical Document #348: Software Engineering - Comprehensive Overview

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Domain-driven design (DDD) models software around business domains. It improves communication between technical and business stakeholders.

Continuous deployment (CD) automatically deploys code changes to production. It enables rapid, reliable releases.

Microservices architecture structures applications as independently deployable services. Each service owns its data and business logic.

Test-driven development (TDD) writes tests before writing code. It improves code quality and design through small, incremental changes.

Cybersecurity threats continue to evolve in complexity and scale. Organizations must invest in robust security measures to protect sensitive data.

Remote work technologies have transformed the workplace. Collaboration tools and cloud services enable distributed teams to work effectively from anywhere.

Cloud computing has democratized access to powerful computing resources. Startups can now access the same infrastructure as large enterprises.

Key Takeaways:

- Microservices architecture structures applications as independently deployable services.
- Design patterns provide reusable solutions to common software design problems.
- Code review examines code changes before merging.